Sri Chaitanya IIT Academy.,India.

A.P T.S KARNATAKA TAMILNADU MAHARASTRA DELHI RANCHI

A right Choice for the Real Aspirant

ICON Central Office - Madhapur - Hyderabad

SEC: Sr.S60_Nu-XL-500 & ALL	JEE-MAIN	Date: 01-01-2026
Time: 7.30AM to 10.30AM	IGTM-04	Max. Marks:300

IMPORTANT INSTRUCTION:

1. Immediately fill in the Admission number on this page of the Test Booklet with **Blue/Black** Ball Point **Pen** only.
2. The candidates should not write their Admission Number anywhere (except in the specified space) on the Test Booklet/ Answer Sheet.
3. The test is of **3 hours** duration.
4. The Test Booklet consists of **75 Questions.** The maximum marks are **300.**
5. There are **three** parts in the question paper 1,2,3 consisting of **Mathematics, Physics** and **Chemistry** having **25 Questions** in each subject and subject having **two sections.**
   (I) Section –I contains **20 Multiple Choice Questions** with only one correct option.
   Marking scheme: +4 for correct answer, **0** if not attempt and **-1** in all other cases.
   (II) Section-II contains **05 Numerical Value Type Questions.**
   - The Answer should be within **0 to 9999.** If the Answer is in **Decimal** then round off to the **Nearest Integer** value (Example i,e. If answer is above **10** and less than **10.5** round off is **10** and If answer is from **10.5** and less than **11** round off is **11).**

   To cancel any attempted question bubble on the question number box.
   For example: To cancel attempted Question 21. Bubble on 21 as shown below

Question Answered for Marking **Question Cancelled for Marking**

Marking scheme: +4 for correct answer, **0** if **not attempt** and **-1** in all other cases.

6. Use **Blue / Black Point Pen only** for writing particulars / marking responses on the Answer Sheet. **Use of pencil is strictly prohibited.**
7. No candidate is allowed to carry any textual material, printed or written, bits of papers, mobile phone any electron device etc, except the Identity Card inside the examination hall.
8. Rough work is to be done on the space provided for this purpose in the Test Booklet only.
9. On completion of the test, the candidate must hand over the Answer Sheet to the invigilator on duty in the Hall. **However, the candidate are allowed to take away this Test Booklet with them.**
10. **Do not fold of make any stray marks on the Answer Sheet**

Name of the Candidate (in Capital): _____

Admission Number: | | | | | | | | | |

Candidate's Signature:_____ **Invigilator's Signature:** _____

01-01-2026_ Sr.Super60_Nu_XL-500 & ALL_Jee-Main_IGTM-04_Test Syllabus

MATHEMATICS	:	TOTAL SYLLABUS
PHYSICS	:	TOTAL SYLLABUS
CHEMISTRY	:	TOTAL SYLLABUS

MATHEMATICS

Max Marks: 100

SECTION-I (SINGLE CORRECT ANSWER TYPE)

This section contains **20 Multiple Choice Questions.** Each question has 4 options (1), (2), (3) and (4) for its answer, out of which ONLY ONE option can be correct.

Marking scheme: +4 for correct answer, 0 if not attempted and –1 in all other cases.

1. Let the triangle PQR be the image of the triangle with vertices (1,3), (3,1), (2,4) in the line $x+2y=2$. If the centroid of ΔPQR is the point (α,β), then $15(\alpha-\beta)$ is equal to ___.

1) 24 2) 19 3) 22 4) 21

2. From all the English alphabets, five letters are chosen and are arranged in alphabetical order. The total number of ways, in which the middle letter M, is

1) 14950 2) 6084 3) 5148 4) 4356

3. Using the principle values of the inverse trigonometric functions, the sum of maximum and minimum values of $16\left(\left(\sec^{-1}x\right)^2+\left(\operatorname{cosec}^{-1}x\right)^2\right)$ is :

1) $31\pi^2$ 2) $18\pi^2$ 3) $22\pi^2$ 4) $24\pi^2$

4. Let for $f(x)=7\tan^8 x+7\tan^6 x-3\tan^4 x-3\tan^2 x$, $I_1=\int_0^{\pi/4} f(x)dx$ and $I_2=\int_0^{\pi/4} x.f(x)dx$, then $7I_1+12I_2$ is equal to

1) π 2) 1 3) 2 4) 2π

5. Let $f:R\to R$ be a twice differentiable function such that $f(x+y)=f(x)f(y)$ for all $x,y\in R$. If $f'(0)=4a$ and f satisfies $f''(x)-3af'(x)-f(x)=0, a>0$, then the area of the region $R=\{(x,y)\,|\,0\le y\le f(ax), 0\le x\le 2\}$ is

1) e^2+1 2) e^4-1 3) e^4+1 4) e^2-1

6. Let $A=\{1,2,3,4,5,6,7,8,9,10\}$, $B=\{m/n : m,n\in A, m<n \text{ and } \gcd(m,n)=1\}$. then $n(B)$ is equal to

1) 37 2) 31 3) 36 4) 29

7. The area of the region, inside the circle $\left(x-2\sqrt{3}\right)^2+y^2=12$ and outside the parabola $y^2=2\sqrt{3}x$ is

1) $6\pi-16$ 2) $3\pi+8$ 3) $3\pi-8$ 4) $6\pi-8$

8. Let $f(x)$ be a real differentiable function such that $f(0)=1$ and $f(x+y)=f(x)f'(y)+f'(x)f(y)$ for all $x,y\in R$, then $\sum_{n=1}^{100}\log_e f(n)$ is equal to.

1) 5220 2) 2525 3) 2384 4) 2406

9. The Product of all solutions of the equation $e^{5(\log_e x)^2+3} = x^8, x>0$ is

1) $e^{6/5}$ 2) e^2 3) $e^{8/5}$ 4) e

10. Let $x = x(y)$ be the solution of the differential equation $y^2dx + \left(x - \frac{1}{y}\right)dy = 0$. If $x(1) = 1$, then $x\left(\frac{1}{2}\right)$ is

1) $3+e$ 2) $\frac{3}{2}+e$ 3) $3-e$ 4) $\frac{1}{2}+e$

11. Let the foci of a hyperbola be (1, 14) and (1, -12). If it passes through the point (1,6), then the length of its latus-rectum is:

1) $\frac{145}{5}$ 2) $\frac{25}{6}$ 3) $\frac{288}{5}$ 4) $\frac{24}{5}$

12. Let z_1, z_2 and z_3 be three complex numbers on the circle $|z| = 1$ with $\arg(z_1) = \frac{-\pi}{4}$, $\arg(z_2) = 0$ and $\arg(z_3) = \frac{\pi}{4}$. If $\left|z_1\overline{z_2} + z_2\overline{z_3} + z_3\overline{z_1}\right|^2 = \alpha + \beta\sqrt{2}$, $\alpha, \beta \in Z$, then the value of $\alpha^2 + \beta^2$ is

1) 29 2) 41 3) 31 4) 24

13. Let $a_1, a_2, a_3, ...$ be a GP of increasing positive terms. if $a_1a_5 = 28$ and $a_2 + a_4 = 29$ then a_6 is

1) 526 2) 812 3) 628 4) 784

14. A coin is tossed three times. Let X denote the number of times a tail follows a head. If μ and σ^2 denote the mean and variance of X, then the value of $64\left(\mu + \sigma^2\right)$ is

1) 48 2) 64 3) 32 4) 51

15. Two balls are selected at random one by one without replacement from a bag containing 4 white and 6 black balls. If the probability that the first selected ball is black, given that the second selected ball is also black, is m/n, where $\gcd(m,n) = 1$, then $m+n$ is equal to

1) 14 2) 11 3) 13 4) 4

16. A circle C of radius 2 lies in the second quadrant and touches both the coordinate axes. Let *r* be the radius of a circle that has centre at the point (2, 5) and intersects the circle C at exactly two points. If the set of all possible values of r is the interval (α, β), then $3\beta - 2\alpha$ is equal to

1) 10 2) 12 3) 15 4) 14

17. Let $L_1 : \frac{x-1}{2} = \frac{y-2}{3} = \frac{z-3}{4}$ and $L_2 : \frac{x-2}{3} = \frac{y-4}{4} = \frac{z-5}{5}$ be two lines. Then which of the following points lies on the line of the shortest distance between L_1 and L_2?

1) $\left(2,3,\frac{1}{3}\right)$ 2) $\left(\frac{14}{3},-3,\frac{22}{3}\right)$ 3) $\left(\frac{8}{3},-1,\frac{1}{3}\right)$ 4) $\left(-\frac{5}{3},-7,1\right)$

18. If $\sum_{r=1}^{n} T_r = \frac{(2n-1)(2n+1)(2n+3)(2n+5)}{64}$, then $\lim_{n\to\infty} \sum_{r=1}^{n} \left(\frac{1}{T_r}\right)$ is equal to :

1) $\frac{2}{3}$ 2) 1 3) $\frac{1}{3}$ 4) 0

19. Let the parabola $y = x^2 + px - 3$, meet the coordinate axes at the points P, Q and R. If the circle C with centre at (–1,-1) passes through the points P, Q and R, then the area of ΔPQR is

1) 7 2) 6 3) 5 4) 4

20. The number of non empty equivalence relations on the set $\{1,2,3\}$ is

1) 7 2) 6 3) 5 4) 4

SECTION-II (NUMERICAL VALUE TYPE)

This section contains **5 Numerical Value Type Questions.** The Answer should be within **0 to 9999.** If the Answer is in **Decimal** then round off to the **Nearest Integer** value (Example i,e. If answer is above **10** and less than **10.5** round off is **10** and If answer is from **10.5** and less than **11** round off is **11**).

Marking scheme: +4 for correct answer, 0 if not attempt and -1 in all other cases

21. Let the function

$$f(x) = \begin{cases} -3ax^2 - 2, & x < 1 \\ a^2 + bx, & x \geq 1 \end{cases}$$

be differentiable for all $x \in R$, where $a > 1, b \in R$. If the area of the region enclosed by $y = f(x)$ and the line $y = -20$ is $\alpha + \beta\sqrt{3}$, $\alpha, \beta \in Z$, then the value of $\alpha + \beta$ is _____.

22. Let A be a square matrix of order 3 such that $\det(A) = -2$ and $\det\left(3adj\left(-6adj\left(3A\right)\right)\right) = 2^{m+n}.3^{mn}$ $m > n$ then $4m + 2n$ is equal to _____.

23. let $L_1 : \frac{x-1}{3} = \frac{y-1}{-1} = \frac{z+1}{0}$ and $L_2 : \frac{x-2}{2} = \frac{y}{0} = \frac{z+4}{\alpha}$, $\alpha \in R$, be two lines which intersect at the point B. If P is the foot of perpendicular from the point A(1,1-1) on L_2, then the value of $26\alpha(PB)^2$ is _____.

24. If $\sum_{r=0}^{5} \frac{{}^{11}C_{2r+1}}{2r+2} = \frac{m}{n}$, $\gcd(m, n) = 1$, then $m - n$ is equal to _____.

25. Let $\vec{c}$ be the projection vector of $\vec{b} = \lambda\hat{i} + 4\hat{k}, \lambda > 0$, on the vector $\vec{a} = \hat{i} + 2\hat{j} + 2\hat{k}$. If $\left|\vec{a} + \vec{c}\right| = 7$, then the area of the parallelogram formed by the vectors $\vec{b}$ and $\vec{c}$ is _____

PHYSICS **Max Marks: 100**

(SINGLE CORRECT ANSWER TYPE)

This section contains 20 multiple choice questions. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which ONLY ONE option can be correct.

Marking scheme: +4 for correct answer, 0 if not attempted and –1 in all other cases.

26. A force $\vec{F} = 2\hat{i} + b\hat{j} + \hat{k}$ is applied on a particle and it undergoes a displacement $\hat{i} - 2\hat{j} - \hat{k}$. What will be the value of b, if work done on the particle is zero.

1) 2 2) 0 3) $\frac{1}{2}$ 4) $\frac{1}{3}$

27. Given below are two statements. One is labeled as Assertion (A) and the other is labeled as Reason (R) .

Assertion (A) : A simple pendulum is taken to a planet of mass and radius, 4 times and 2 times, respectively, then the Earth . The time period of the pendulum remains same on earth and the planet .

Reason (R) : The mass of the pendulum remains unchanged at earth and the other planet . In the light of the above statements , choose the correct answer from the options given below :

1) Both (A) and (R) are true but (R) is NOT the correct explanation of (A)
2) (A) is false but (R) is true
3) Both (A) and (R) are true and (R) is the correct explanation of (A)
4) (A) is true but (R) is false

28. The torque due to the force $\left(2\hat{i} + \hat{j} + 2\hat{k}\right)$ about the origin, acting on a particle whose position vector is $\left(\hat{i} + \hat{j} + \hat{k}\right)$, would be

1) $\hat{j} + \hat{k}$ 2) $\hat{i} - \hat{k}$ 3) $\left(\hat{i} - \hat{i} + \hat{k}\right)$ 4) $\hat{i} + \hat{k}$

29. For a diatomic gas, if $\gamma_1 = \left(\frac{C_p}{C_v}\right)$ for rigid molecules and $\gamma_2 = \left(\frac{C_p}{C_v}\right)$ for another diatomic molecules, but also having vibrational modes, Then, which one of the following options is correct ?

(C_p and C_v are specific heats of the gas at constant pressure and volume

1) $\gamma_2 = \gamma_1$ 2) $\gamma_2 > \gamma_1$ 3) $2\gamma_2 = \gamma_1$ 4) $\gamma_2 < \gamma_1$

30. A transparent film of refractive index, 2.0 is coated on a glass slab of refractive index, 1.45. What is the minimum thickness of transparent film to be coated for the maximum transmission of Green light of wavelength 550 nm. [Assume that the light is incident nearly perpendicular to the glass surface.]

1) 68.7 nm 2)275 nm 3) 94.8 nm 4) 137.5

31. Given are statements for certain thermodynamic variables,

A) Internal energy, volume (V) and mass (M) are extensive variables.

B) Pressure (P) , temperature (T) and density (p) are intensive variables .

C) Volume (V), temperature (T) and density (p) are intensive variables.

D) Mass (M), temperature (T) and internal energy are extensive variables.

Choose the correct answer from the options given below :

1) (D) and (A)Only 2) (C) and (D) Only

3) (A) and (B) Only 4) (B) and (C) Only

32. Which one of the following is the correct dimensional formula for the capacitance in F ?

M. L, T and C stand for unit of mass, length time and charge ,

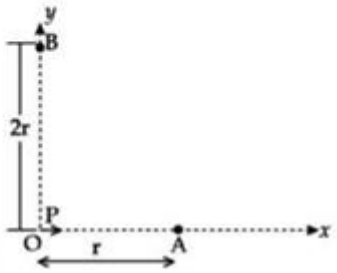
1) $[F]=\left[C^2M^{-2}L^2T^2\right]$ 2) $[F]=\left[C^2M^{-1}L^{-2}T^2\right]$

3) $[F]=\left[CM^{-2}L^{-2}T^{-2}\right]$ 4) $[F]=\left[CM^{-1}L^{-2}T^2\right]$

33. A ball of mass 100 g is projected with velocity 20 m/s at 60^0 with horizontal. The decrease in kinetic energy of the ball during the motion from point of projection to highest point is

1) 15J 2) 20J 3) Zero 4) 5J

34. For a short dipole placed at origin O, the dipole moment P is along x-axis , as shown in the figure. If the electric potential and electric field at A are V_0 and E_0 , respectively then the correct combination of the electric potential and electric field, respectively, at point B on the y-axis is given by



1) Zero and $\frac{E_0}{16}$ 2) Zero and $\frac{E_0}{8}$

3) V_0 and $\frac{E_0}{4}$ 4) $\frac{V_0}{2}$ and $\frac{E_0}{16}$

35. Given below are two statements . One is labeled as Assertion (A) and the other is labeled as Reason (R) .

Assertion (A) : In young's double slit experiment, the fringes produced by red light are closer as compared to those produced by blue light.

Reason (R) : The fringe width is directly proportional to the wavelength of light. In the light of the above statements, choose the correct answer to the options given below :

1) Both (A) and (R) are true and (R) is the correct explanation of (A)
2) (A) is false but (R) is true
3) (A) is true but (R) is false
4) Both (A) and (R) are true but (R) is NOT the correct explanation of (A)

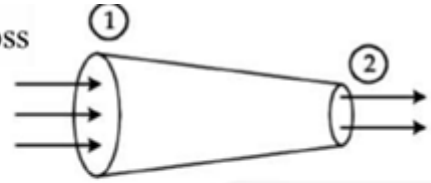
36. An electron projected perpendicular to a uniform magnetic field B moves in a circle. If Bohr's quantization is applicable, then the radius of the electronic orbit in the first excited state is :

1) $\sqrt{\frac{h}{2\pi eB}}$ 2) $\sqrt{\frac{2h}{\pi eB}}$ 3) $\sqrt{\frac{h}{\pi eB}}$ 4) $\sqrt{\frac{4h}{\pi eB}}$

37. A series LCR circuit is connected to an alternating source of emf E. The current amplitude at resonant frequency is I_0. If the value of resistance R becomes twice of its initial value then amplitude of current at resonance will be

1) $2I_0$ 2) $\frac{I_0}{2}$ 3) I_0 4) $\frac{I_0}{\sqrt{2}}$

38. A tube of length L is shown in the figure. The radius of cross section at the point (1) is 2 cm and at the point (2) is 1 cm, respectively. If the velocity of water entering at point (1) is 2 m/s, then velocity of water leaving the point (2) will be

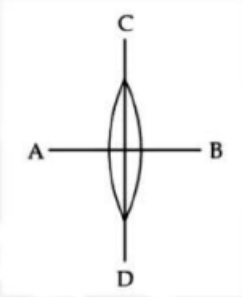


1)8 m/s 2) 4 m/s 3) 6 m/s 4) 2 m/s

39. The time period of a body undergoing simple harmonic motion is given by $T = p^a D^b S^c$, where p is pressure, D is density, and S is surface tension. The values of a, b and c respectively are

1) $1, \frac{1}{2}, \frac{3}{2}$ 2) $\frac{3}{2}, -\frac{1}{2}, 1$ 3) $1, -\frac{1}{2}, \frac{3}{2}$ 4) $-\frac{3}{2}, \frac{1}{2}, 1$

40. A symmetric thin biconvex lens is cut into four equal parts by two panes AB and CD as shown in figure. If the power of original lens is 4D then the power of a part of the divided lens is



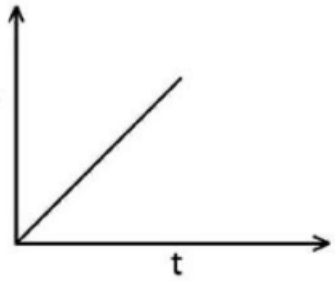
1) 8D 2) 2D 3) D 4) 4D

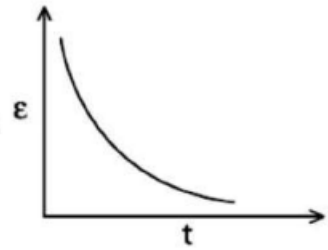
41. The maximum percentage error in the measurement of density of wire is

$$\left[\begin{array}{ll}\text{Given,} & \text{mass of wire} = (0.60 \pm 0.003)\,\text{g} \\ & \text{radius of wire} = (0.50 \pm 0.01)\,\text{cm} \\ & \text{length of wire} = (10.00 \pm 0.05)\,\text{cm}\end{array}\right]$$

1) 5 2) 7 3) 8 4) 4

42. A rectangular metallic loop is moving out of uniform magnetic field region to a field region with a constant speed. When the loop is partially inside the magnate field, the plot of magnitude of induced emf (ε) with time (t) is given by

1.


2.


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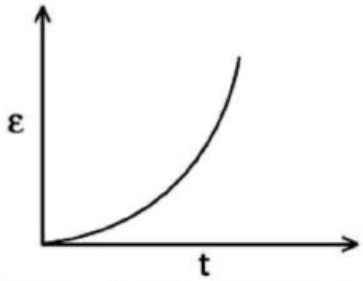
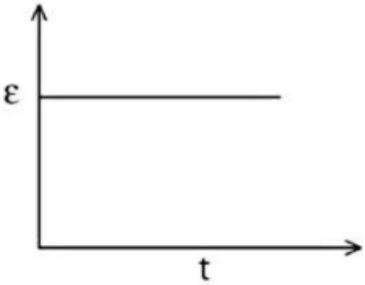
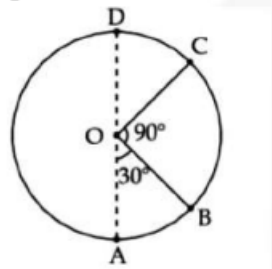


4.


43. A light source of wavelength λ illuminates a metal surface and electrons are ejected with maximum kinetic energy of 2 eV. If the same surface is illuminated by a light source of wavelength $\frac{\lambda}{2}$, then the maximum kinetic energy of ejected electrons will be (The work function of metal is 1 eV)

1) 5 eV 2) 6 eV 3) 2 eV 4) 3eV

44. A body of mass 100g is moving in circular path of radius 2 m on vertical plane as shown in figure. The velocity of the body at point A is 10 m/s . The ratio of its kinetic energies at point B and C is :



(Take acceleration due to gravity as 10 m/s^2)

1) $\frac{2+\sqrt{3}}{3}$ 2) $\frac{3-\sqrt{2}}{2}$ 3) $\frac{2+\sqrt{2}}{3}$ 4) $\frac{3+\sqrt{3}}{2}$

45. A small rigid spherical ball of mass M is dropped in a long vertical tube containing glycerine . The velocity of the ball becomes constant after some time. If the density of glycerine is half of the (consider g as acceleration due to gravity)

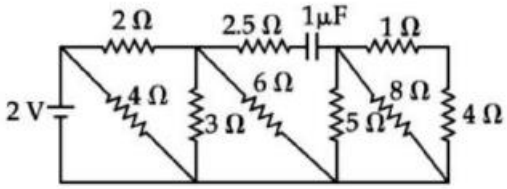
1) $\frac{Mg}{2}$ 2)Mg 3) $\frac{3}{2}Mg$ 4) 2 Mg

SECTION-II (NUMERICAL VALUE TYPE)

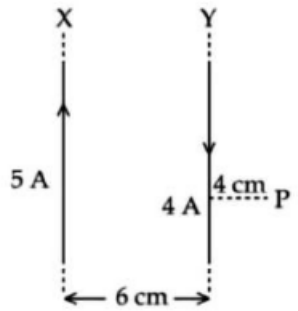
This section contains **5 Numerical Value Type Questions.** The Answer should be within **0 to 9999.** If the Answer is in **Decimal** then round off to the **Nearest Integer** value (Example i,e. If answer is above **10** and less than **10.5** round off is **10** and If answer is from **10.5** and less than **11** round off is **11**).

Marking scheme: +4 for correct answer, 0 if not attempt and -1 in all other cases

46. The net current flowing in the given circuit is _____A .



47. Two long parallel wires X and Y, separated by a distance of 6 cm, carry currents of 5A and 4 A, respectively, in opposite direction as shown in the figure. Magnitude of the resultant magnetic field at point P at a distance of 4 cm from wire Y is $x\times10^{-5}T$. The value of x is _____ . Take permeability of free space as $\mu_0 = 4\pi\times10^{-7}SI$ units .



48. A parallel plate capacitor of area $A=16cm^2$ and separation between the plates 10 cm, is charged by a DC current. Consider a hypothetical plane surface of area $A_0 = 3.2\ Cm^2$ inside the capacitor and parallel to the plates. At an instant, the current through the circuit is 6A. At the same instant the displacement current through A_0 is _____mA.

49. A tube of length 1m is filled completely with an ideal liquid of mass 2 M, and close at both ends. The tube is rotated uniformly in horizontal plane about one of its ends. If the force exerted by the liquid at the other end is F the angular velocity of the tube is $\sqrt{\frac{F}{\alpha M}}$ in SI unit. The value of α is _____

50. A proton is moving undeflected in a region of crossed electric and magnetic fields at a constant speed of $2x10^5 ms^{-1}$. When the electric field is switched off, the proton moves along a circular path of radius 2 cm. The magnitude of electric field $x\times10^4\ N/C$. The value of x is _____

Take the mass of proton $=1.6x10^{-27}kg$.

CHEMISTRY

Max Marks: 100

SECTION-I (SINGLE CORRECT ANSWER TYPE)

This section contains **20 Multiple Choice Questions.** Each question has 4 options (1), (2), (3) and (4) for its answer, out of which ONLY ONE option can be correct.

Marking scheme: +4 for correct answer, 0 if not attempted and –1 in all other cases.

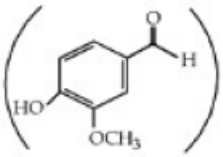
51. If equal volumes of AB_2 and XY (both are salts) aqueous solutions are mixed, which of the following Combination will give a precipitate of AY_2 at 300K?
(Given $K_{sp}\ (at\ 300K)$ for $AY_2 = 5.2\times10^{-7}$)

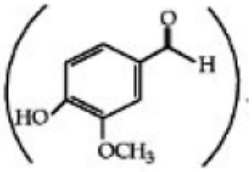
1) $2.0\times10^{-2}M\ AB_2, 2.0\times10^{-2}M\ XY$
2) $3.6\times10^{-3}M\ AB_2, 5.0\times10^{-4}M\ XY$
3) $1.5\times10^{-4}M\ AB_2, 1.5\times10^{-3}M\ XY$
4) $2.0\times10^{-4}M\ AB_2, 0.8\times10^{-3}M\ XY$

52. An optically active alkyl halide $C_4H_9B_r[A]$ reacts with hot KOH dissolved in ethanol and forms alkene $[B]$ as major product which reacts with bromine to give dibromide $[C]$. The compound $[C]$ is converted into a gas $[D]$ upon reacting with alcoholic $NaNH_2$. During hydration 18 gram of water is added to 1 mole of gas $[D]$ on warming with mercuric sulphate and dilute acid and at 333K to from compound [E]. The IUPAC name of compound [E] is :

1) Butan-2-one 2) Butan-1-al 3) But-2-yne 4) Butan-2-ol

53. Given below are two statements :

Statement (I) : Vanillin  will react with $NaOH$ and also with Tollen's reagent

Statement (II) : Vanillin  will undergo self aldol condensation very easily.

In the light of above statements, choose the most appropriate answer from the options given below :

1. Statement I is correct but Statement II is incorrect
2. Statement I is incorrect but Statement II is correct
3. Both Statement I and Statement II are correct
4. Both Statement I and Statement II are incorrect

54. Among $SO_2, NF_3, NH_3, XeF_2, ClF_3$ and SF_4, the hybridization of the molecule with non-zero dipole moment and highest number of lone-pairs of electrons on the central atom is

1) sp^3d^2 2) sp^3d 3) dsp^2 4) sp^3

55. Choose the correct tests with respectively observations:

A) $CuSO_4$(acidified with acetic acid)+ $K_4\left[Fe(CN)_6\right] \rightarrow$ Chocolate brown precipitate

B) $FeCl_3 + K_4\left[Fe(CN)_6\right] \rightarrow$ Prussian blue precipitate

C) $ZnCl_2 + K_4\left(Fe(CN)_6\right)$ neutralised with $NH_4OH \rightarrow$ white or bluish white precipitate

D) $MgCl_2 + K_4\left[Fe(CN)_6\right] \rightarrow$ Blue precipitate

E) $BaCl_2 + K_4\left[Fe(CN)_6\right]$ neutralised with $NaOH \rightarrow$ White precipitate.

Choose the correct answer from the options given below:

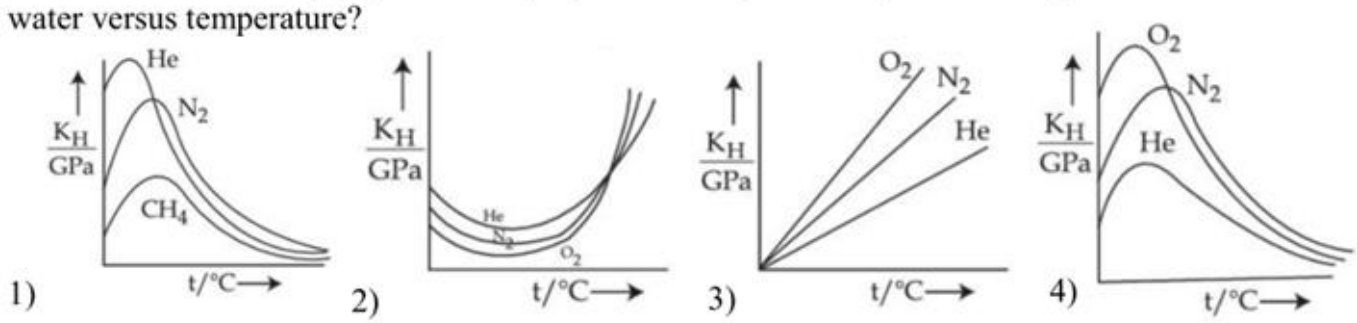
1) C, D and E only 2) A, B and C only
3) A, D and E only 4) B, D and E only

56. Given below are two statements :
Statement –I : The metallic radius of Al is less than that of Ga.
Statement –II : The ionic radius of Al^{3+} is less than that of Ga^{3+}
In the light of the above statements, choose the most appropriate answer from the options given below.
1. Both Statement I and Statement II are incorrect
2. Statement I is correct but Statement II is incorrect
3. Both Statement I and Statement II are correct
4. Statement I is incorrect but Statement II is correct

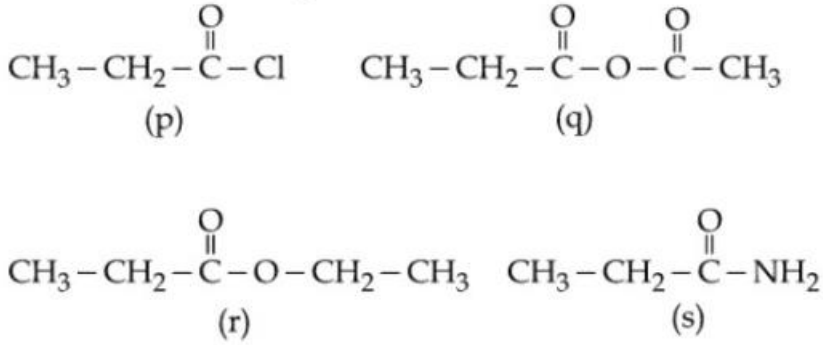
57. A solution made by mixing one mole of volatile liquid A with 3 moles of volatile liquid B. The vapour pressure of pure A is 200 mm Hg and that of the solution in 500 mm Hg. The vapour pressure of pure B and the least volatile component of the solution, respectively are :
1) 600 mm Hg, B
2) 1400 mm Hg, A
3) 1400 mm Hg, B
D) 600 mm Hg, A

58. On complete combustion 1.0g of an organic compound (X) gave 1.46 g of CO_2 and 0.567 g of H_2O. The empirical formula mass of compound (X) is _____g.
(Given molecular mas in g $mol^{-1}C$: 12,H:1,O:16)
1) 30
2) 15
3) 60
4) 45

59. Which of the following graph correctly represents the plots of K_H at 1 bar for gases in water versus temperature?



60. Consider the following molecules :



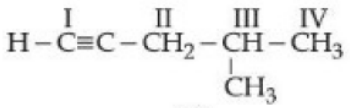
The correct order of rate of hydrolysis is :
1) $r > q > p > s$
2) $q > p > r > s$
3) $p > r > q > s$
4) $p > q > r > s$

61. Identify the correct statement among the following :
1) All naturally occurring amino acids except glycine contains one chiral centre
2) All naturally occurring amino acids are optically active.
3) Glutamic acid the only amino acid that contains a –COOH group at the side chain.
4) Amino acid, cysteine can easily undergoes dimerization due to the presence of free SH group.

62. The property/properties that show irregularity in first elements of group-17 is/are :
1) Covalent radius 2) Electron affinity
3) Ionic radius 4) First ionization energy
Choose the correct answer from the options given below :
1) B only 2) A and C only 3) A,B,C and D 4) B and D only

63. Given below are two statements :
Statement –I : In octahedral complexes, when $\Delta_0 < P$ high spin complexes are formed. When $\Delta_0 > P$ low spin complexes are formed.
Statement – II : In tetrahedral complexes because of $\Delta_t < P$, low spin complexes are rarely formed. In the light of the above statements, choose the most appropriate answer from the options given below :
1) Statement I is correct but Statement II is incorrect
2) Statement I is incorrect but Statement II is correct
3) Both Statement I and Statement II are correct
4) Both Statement I and Statement II are incorrect.

64. Consider the following compound (X)



(X)

The most stable and least stable carbon radicals, respectively, produced by homolytic cleavage of corresponding C-H bond are :
1) I,IV 2) III,II 3) II,IV 4) II,I

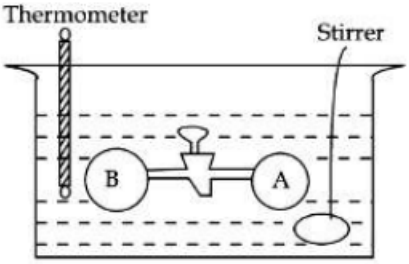
65. The correct order of basic nature in aqueous solution for the bases :
NH_3, H_2N-NH_2, $CH_3CH_2NH_2$, $(CH_3CH_2)_2NH$ and $(CH_3CH_2)_3N$

1. $NH_2-NH_2 < NH_3 < CH_3CH_2NH_2 < (CH_3CH_2)_3N < (CH_3CH_2)_2NH$
2. $H_2N-NH_2 < NH_3 < (CH_3CH_2)_3N < CH_3CH_2NH_2 < (CH_3CH_2)_2NH$
3. $NH_3 < H_2N-NH_2 < CH_3CH_2NH_2 < (CH_3CH_2)_2NH < (CH_3CH_2)_3N$
4. $NH_3 < H_2N-NH_2 < (CH_3CH_2)_3N < CH_3CH_2NH_2 < (CH_3CH_2)_2NH$

66. A molecule with the formula AX_4Y has all its elements from p-block. Element A is rarest monoatomic, non-radioactive from its group and has lowest ionization enthalpy value among A,X and Y. Elements X and Y have first and second highest electronegativity values respectively among all the known elements. The shape of the molecule is :

1) Trigonal bipyramidal
2) Square pyramidal
3) Pentagonal planar
4) Octahedral

67.



The vessels A and B are connected via stopcock. The vessel A is filled with a gas at a certain pressure. The entire assembly is immersed in water and is allowed to come to thermal equilibrium with water. After opening the stopcock the gas from vessel A expands into vessel B and no change in temperature is observed in the thermometer. Which of the following statement is true.

1) $dq \neq 0$
2) $dU \neq 0$
3) $dw \neq 0$
4) The pressure in the vessel B before opening the stopcock is zero.

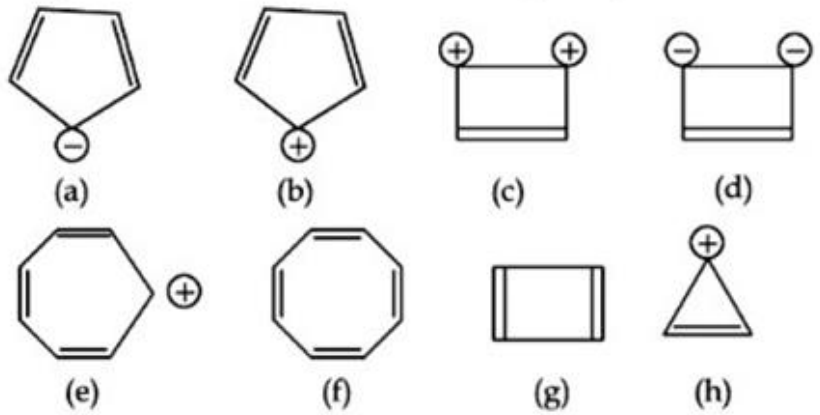
68. $CaCO_3(s) + 2HCl(aq) \rightarrow CaCl_2(aq) + CO_2(g) + H_2O(l)$

Consider the above reaction, what mass of $CaCl_2$ will be formed if 250 ml, of 0.76 M HCl reacts with 1000g of $CaCO_3$?

(Given : Molar mass of Ca, C, O, H and Cl are 40,12,16,1 and 35.5 g mol^{-1} respectively)

1) 10.545 g
2) 3.908 g
3) 5.272 g
4) 2.636 g

69. Designate whether each of the following compounds is aromatic or not aromatic.



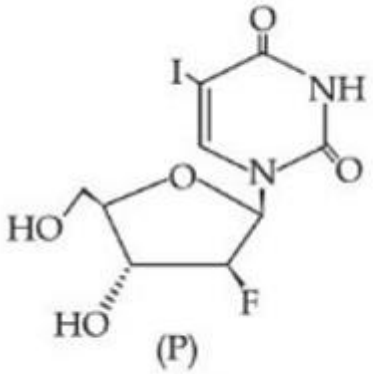
1) a,c,d,e,h aromatic and b,f,g not aromatic
2) e,g aromatic and a,b,c,d,f,h not aromatic
3) b,e,f,g aromatic and a,c,d,h not aromatic
4) a,b,c,d aromatic and e,f,g,h not aromatic

70. According to Bohr's model of hydrogen atom, which of the following statement is incorrect?
A) Radius of 4th orbit is four times larger than that of 2nd orbit
B) Radius of 8th orbit is four times larger than that of 4th orbit
C) Radius of 3rd orbit is nine times larger than that of 1st orbit
D) Radius of 6th orbit is three times larger than that of 4th orbit.

SECTION-II (NUMERICAL VALUE TYPE)

This section contains **5 Numerical Value Type Questions.** The Answer should be within **0 to 9999.** If the Answer is in **Decimal** then round off to the **Nearest Integer** value (Example i,e. If answer is above **10** and less than **10.5** round off is **10** and If answer is from **10.5** and less than **11** round off is **11**).
Marking scheme: +4 for correct answer, 0 if not attempt and -1 in all other cases

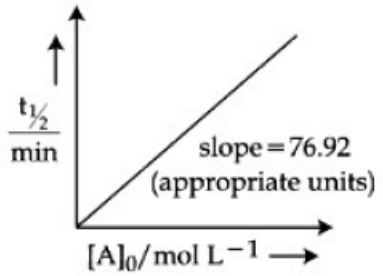
71. 0.1 mol of the following given antiviral compound (P) will weigh _____ $\times 10^{-1} g$



(P)

(Given : molar mass in g $mol^{-1} H:1, C:12, N;14, O:16, F:19, I:127$)

72. For the reaction $A \rightarrow$ products



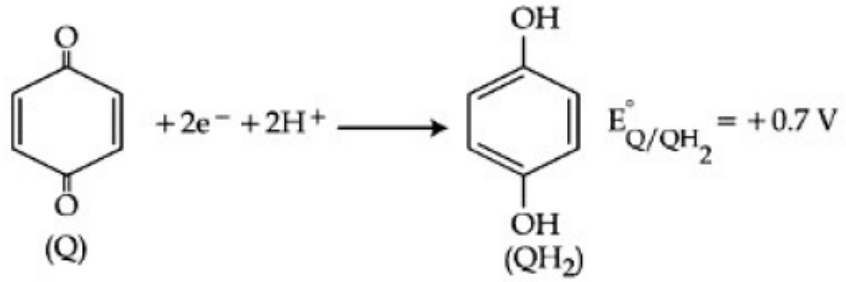
The concentration of A at 10 minutes is _____ $\times 10^{-3} mol\ L^{-1}$ (nearest integer). The reaction was started with 2.5 mol L^{-1} of A.

73. Consider the following equilibrium $CO(g) + 2H_2(g) \rightleftharpoons CH_3OH(g)$
0.1 mol of CO along with a catalyst is present in a 2 dm^3 flask maintained at 500K. Hydrogen is introduced into the flask until the pressure is 5 bar and 0.04 mol of CH_3OH is formed. The K_p^0 is _____ $\times 10^{-3}$ (nearest integer).
Given: $R = 0.08\ dm^3$ bar K^{-1} mol^{-1}
Assume only methanol is formed as the product and the system follows ideal gas behaviour.

74. Consider the following electrochemical cell at standard condition

$Au(s)|QH_2,Q|NH_4X(0.01\ M)||Ag^+(1\ M)|Ag(s)$ $E_{cell} = +0.4\ V$

The couple QH_2/Q represents quinhydrone electrode, the half cell reaction is given below



[Given : $E^0_{Ag^+/Ag} = +0.8V$ and $\frac{2.303RT}{F} = 0.06V$]

The pK_b value of the ammonium halide salt (NH_4X) used here is _____(nearest integer)

75. A transition metal (M) among Mn,Cr,Co and Fe has the highest standard electrode potential (M^{3+}/M^{2+}). If forms a metal complex of the type $[M(CN)_6]^{4-}$. The number of electrons present in the e_g orbital of the complex is _____